

HEATSHIELD: SMART TEMPERATURE MONITORING SYSTEM FOR HEAT STROKE PREVENTION IN INDUSTRIAL WORKERS

PROJECT REPORT

Submitted for

Task 1 – Introduction to Embedded Systems & IoT Device Design

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CHAPTER 1

ABSTRACT

Workers in construction sites, agricultural fields, mining industries, and other outdoor environments often spend long hours under extreme weather conditions. Continuous exposure to high temperatures can lead to dehydration, heat exhaustion, reduced productivity, and in severe cases, heat stroke. Since these conditions can develop gradually, workers may not always recognize the warning signs before their health is affected.

To address this issue, a Smart Temperature Monitoring System named HeatShield was developed using an Arduino Uno microcontroller and a TMP36 temperature sensor. The system continuously monitors environmental temperature and classifies the conditions into safe, warning, and critical levels based on predefined thresholds. Visual and audible alerts are generated using an LED and buzzer whenever unsafe temperature conditions are detected.

The system was designed and tested using the Tinkercad simulation platform. The simulation results showed that the system successfully identified different temperature conditions and generated appropriate warnings in real time. The proposed solution is simple, cost-effective, and suitable for integration into smart safety helmets and wearable safety devices for industrial workers.

This project demonstrates how embedded systems can be used to improve workplace safety by providing early warnings and helping prevent heat-related illnesses.

CHAPTER 2

INTRODUCTION

2.1 Introduction to Embedded Systems

An embedded system is a combination of hardware and software designed to perform a specific task. Unlike a personal computer, which can perform many different functions, an embedded system is developed to carry out a dedicated operation efficiently and reliably.

Embedded systems are present in many devices that we use every day, such as washing machines, microwave ovens, smart watches, automobiles, medical equipment, and industrial machines. These systems usually consist of a microcontroller, sensors, input/output devices, and a program that controls the overall operation.

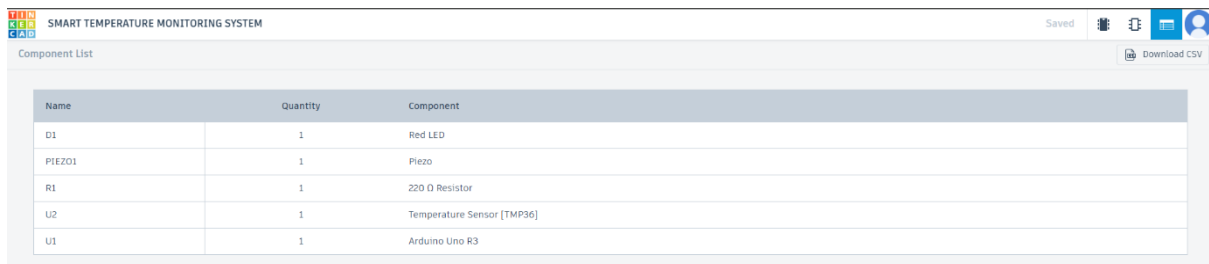
One of the major advantages of embedded systems is their ability to perform tasks automatically with minimal human intervention. Because of their compact size, low power consumption, and cost-effectiveness, they are widely used in modern technology.

2.2 Project Overview

During the study of workplace safety, it was observed that many outdoor workers are regularly exposed to high temperatures. Workers such as construction labours, traffic police officers, farmers, and mining personnel often spend several hours under direct sunlight or in hot environments. Continuous exposure to heat can affect their health and reduce productivity.

In many situations, workers may not realize that the surrounding temperature has reached a dangerous level until symptoms such as dizziness, fatigue, or dehydration begin to appear. This highlights the need for a simple monitoring system that can provide early warnings before conditions become critical.

To address this issue, the Heat-Shield system was developed. The system continuously measures environmental temperature and alerts the user whenever unsafe temperature levels are detected. The goal is to create a low-cost and practical solution that can contribute to worker safety and help prevent heat-related illnesses.



Name	Quantity	Component
D1	1	Red LED
PIEZ01	1	Piezo
R1	1	220 Ohm Resistor
U2	1	Temperature Sensor [TMP36]
U1	1	Arduino Uno R3

Figure 1: Components Used

CHAPTER 3

DESIGN METHODOLOGY

3.1 System Architecture

The Heat-Shield system consists of the following components:

- Arduino Uno
- TMP36 Temperature Sensor
- LED Indicator
- Piezo Buzzer
- Breadboard and Jumper Wires

The TMP36 sensor measures the surrounding temperature and sends the data to the Arduino Uno. The Arduino processes the sensor readings and compares them with predefined temperature limits. Based on the measured temperature, the system decides whether the environment is safe or if a warning should be generated.

The system operates under three conditions:

Safe Condition (Below 35°C)

- LED OFF
- Buzzer OFF

High Temperature Warning (35°C – 40°C)

- LED ON
- Buzzer OFF

Heat Stroke Warning (Above 40°C)

- LED ON
- Buzzer ON

These warning levels help workers become aware of increasing temperature conditions and take preventive measures before their health is affected.

3.2 Circuit Diagram Explanation

The TMP36 temperature sensor is connected to the Arduino Uno through analog pin A0. The sensor continuously measures environmental temperature and sends an analog voltage signal to the microcontroller.

An LED is connected to digital pin D8 through a resistor and acts as a visual indicator. A buzzer is connected to digital pin D9 and is used to provide an audible alert when the temperature reaches a critical level.

The Arduino continuously reads the sensor output, converts it into temperature values, and activates the LED and buzzer according to the programmed threshold conditions.

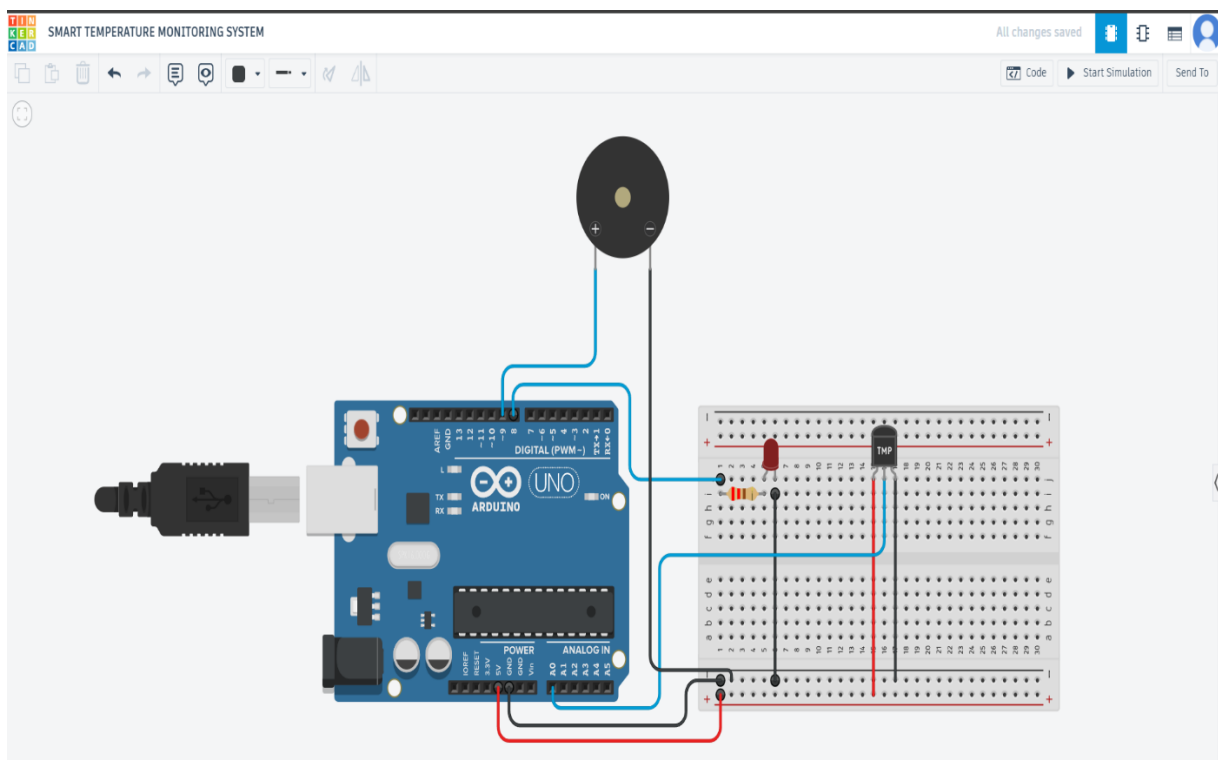


Figure 2: Circuit Connection

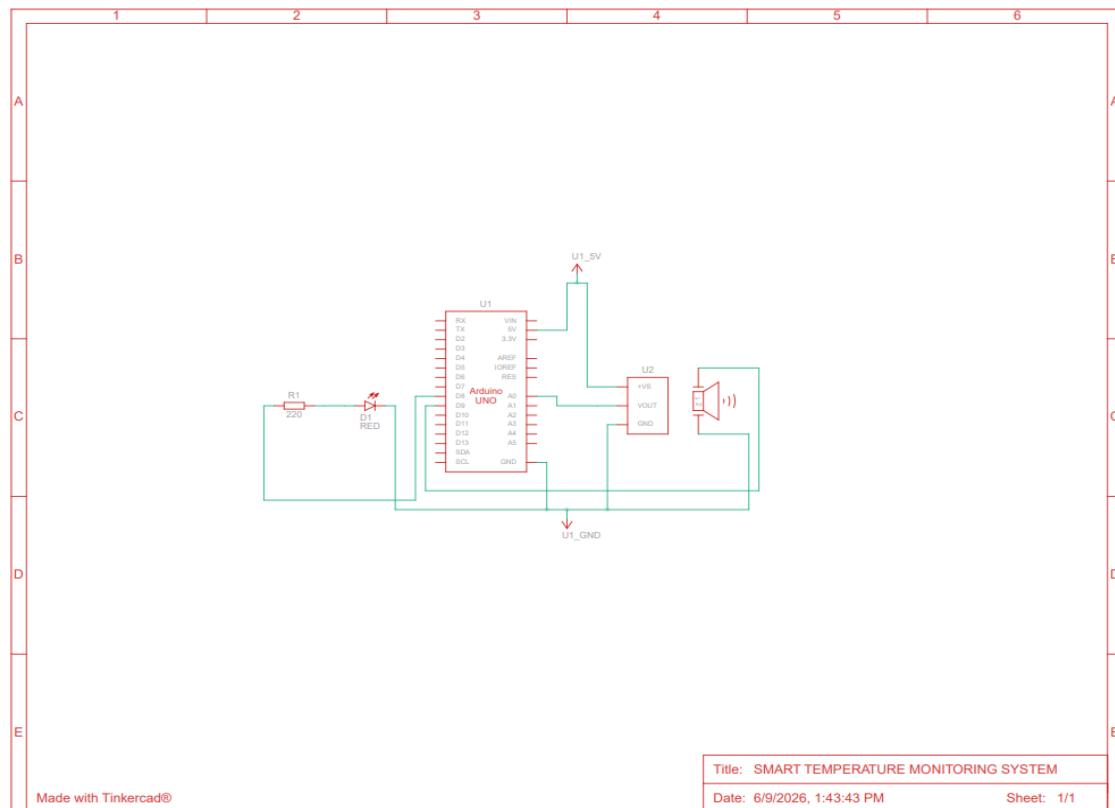


Figure 3: Schematic Diagram

3.3 Working Principle

The working principle of the Heat-Shield system is simple and effective. The Heat-Shield system continuously monitors the surrounding temperature using a TMP36 temperature sensor. The sensor measures the environmental temperature and sends the readings to the Arduino Uno for processing. The Arduino compares the measured temperature with predefined threshold values and determines the current safety condition.

When the temperature is below **35°C**, the environment is considered safe, and both the LED and buzzer remain OFF. If the temperature rises between **35°C and 40°C**, the LED turns ON to indicate a high-temperature warning. When the temperature exceeds **40°C**, the system identifies a potential heat stroke risk and activates both the LED and buzzer to provide an immediate alert.

This continuous monitoring and alert mechanism help workers become aware of unsafe temperature conditions and take preventive action before heat-related illnesses occur.

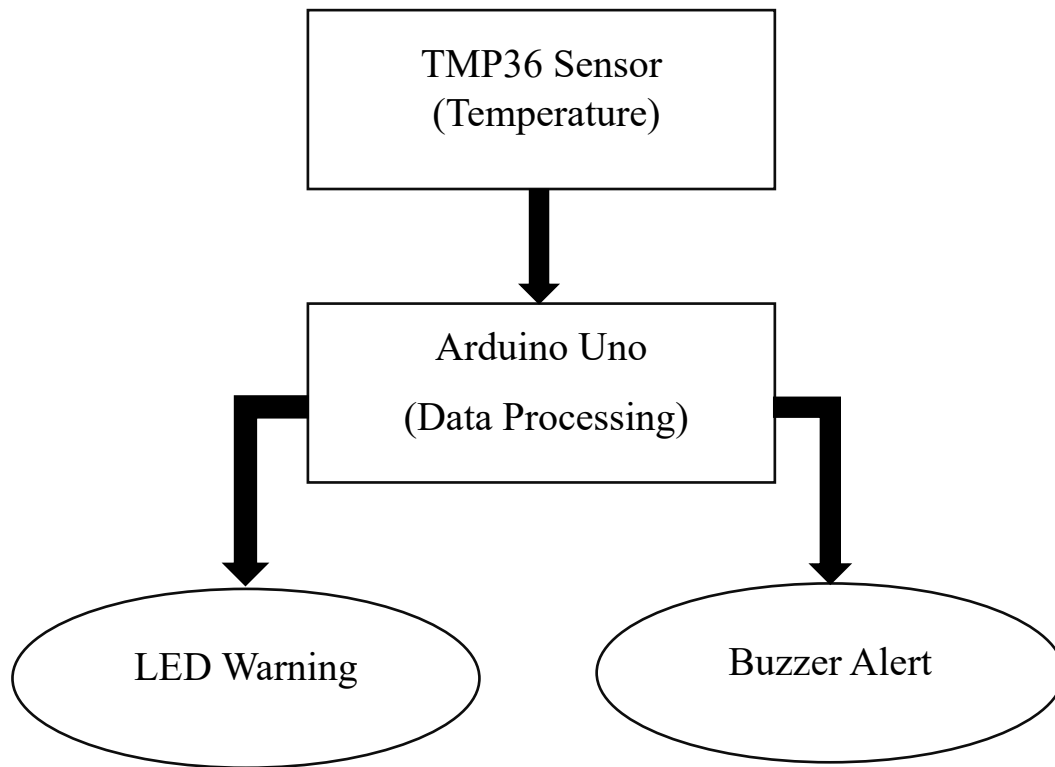


Figure 4: Block Diagram of Heat-Shield Smart Temperature Monitoring System

CHAPTER 4

RESULTS

The Heat-Shield system was successfully designed and tested using the Tinkercad simulation platform. Different temperature conditions were simulated to verify the performance of the system.

4.1 Safe Condition

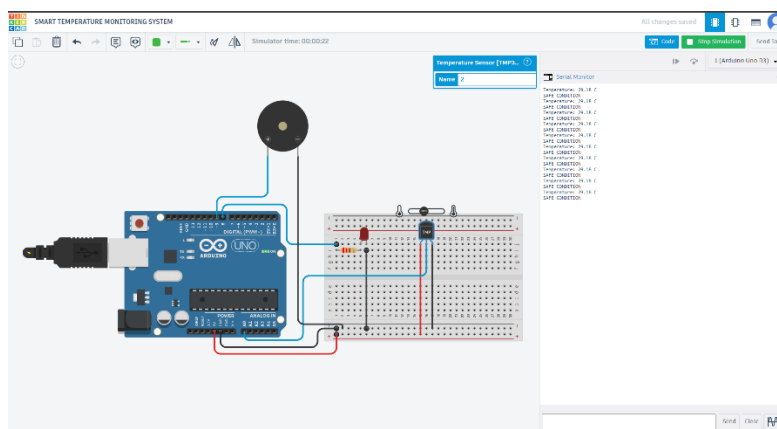


Figure 5: Safe Condition

Observation:

When the temperature was maintained below 35°C, the system operated normally. The LED and buzzer remained OFF, indicating that the environment was safe for the worker.

4.2 High Temperature Warning

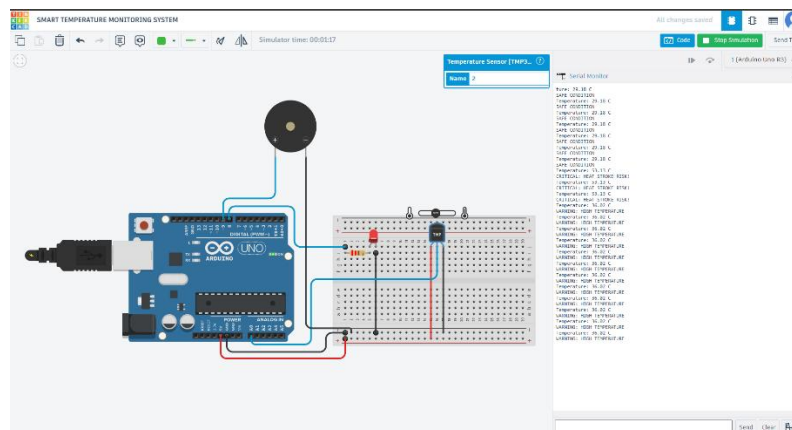


Figure 6: High Temperature Warning

Observation:

When the temperature increased above 35°C, the LED turned ON and a warning message was displayed. This condition indicates that the worker is operating in an elevated temperature environment and should take precautions.

4.3 Heat Stroke Warning

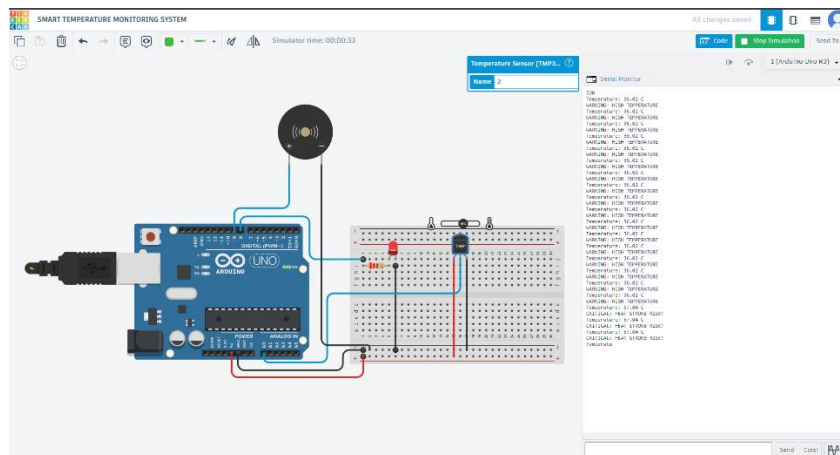


Figure 7: Heat Stroke Warning

Observation

When the temperature exceeded 40°C, both the LED and buzzer were activated. This alert indicates a potentially dangerous condition and warns the worker about the risk of heat stroke.

4.4 Real-World Applications

The Heat-Shield system can be used in several real-world applications, including:

- Construction worker safety monitoring
- Smart safety helmets for industrial workers
- Traffic police heat exposure monitoring
- Agricultural field worker protection
- Mining and manufacturing industries
- Outdoor workforce safety systems

By providing early temperature warnings, the system can help reduce heat-related health risks and improve worker safety.

CHAPTER 5

CONCLUSION

In this project, a Smart Temperature Monitoring System called Heat-Shield was developed using an Arduino Uno and a TMP36 temperature sensor. The objective was to create a simple and effective solution that can help workers identify unsafe temperature conditions before they become harmful.

The simulation results showed that the system was able to successfully classify temperature conditions into safe, warning, and critical levels. Appropriate alerts were generated using an LED and buzzer whenever the temperature exceeded predefined thresholds.

This project demonstrates how embedded systems can be used to solve practical real-world problems. Although the current system was tested through simulation, it can be further developed into a wearable safety device or integrated into industrial safety helmets for real-time monitoring. Overall, the project highlights the importance of technology in improving workplace safety and preventing heat-related illnesses.

REFERENCES

1. Arduino Official Documentation, <https://www.arduino.cc>
2. TMP36 Temperature Sensor Datasheet, Analog Devices.
3. Tinkercad Circuits Simulator, <https://www.tinkercad.com>
4. NPTEL – Introduction to Embedded Systems.
5. Arduino Uno Technical Specifications.

ANNEXURE 1 (PROGRAM)

Arduino Source Code

```
// Heat-Shield: Smart Temperature Monitoring System
// Developed for Heat Stroke Prevention in Industrial Workers

const int tempPin = A0;    // TMP36 sensor connected to A0
const int ledPin = 8;      // LED connected to D8
const int buzzerPin = 9;   // Buzzer connected to D9

void setup()
{
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  // Read temperature sensor value
  int reading = analogRead(tempPin);

  // Convert sensor reading into voltage
  float voltage = reading * (5.0 / 1023.0);

  // Convert voltage into temperature (°C)
  float temperatureC = (voltage - 0.5) * 100.0;

  // Display temperature on Serial Monitor
  Serial.print("Temperature: ");
  Serial.print(temperatureC);
  Serial.println(" °C");
}
```

```
// Critical Condition: Heat Stroke Warning
if (temperatureC >= 40)
{
    digitalWrite(ledPin, HIGH);
    tone(buzzerPin, 1000);
    Serial.println("HEAT STROKE WARNING");
}

// High Temperature Warning
else if (temperatureC >= 35)
{
    digitalWrite(ledPin, HIGH);
    noTone(buzzerPin);
    Serial.println("HIGH TEMPERATURE WARNING");
}

// Safe Condition
else
{
    digitalWrite(ledPin, LOW);
    noTone(buzzerPin);
    Serial.println("SAFE CONDITION");
}
delay(1000);
}
```